REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	ECN00239 - APPROVED 11/19/2020	1/27/2021	A. NICHOLS
	B	ECN00336 - MOD APPROVED 9/21/2021	9/29/2021	A. NICHOLS

UL Color Codes

Abbreviation	Color
AL	Aluminum
ALL	Any color
AM	Amber
AO	All opaque
AT	All transparent
BG	Beige
BK	Black
BL	Blue
BN	Brown
BZ	Bronze
CH	Charcoal
CL	Clear
CT	Clear transparent
DK	Dark
GD	Gold
GN	Green
GY	Gray
GT	Granite
IV	Ivory
LT	Light
NC	Natural (no pigmentation)
OL	Olive
OP	Opaque
OR	Orange
PK	Pink
RD	Red
SM	Smoke
TL	Translucent
TN	Tan
TP	Transparent
VT	Violet
WT	White
YL	Yellow

Unless otherwise noted, the following specifications must be followed.

Wire Specifications:

- **Mil-Spec:** MIL-W-16878/4
- **RoHS:** Not Required

Example: Belden 83007

P: Power Cable
R: RF Cable
C: Control/Command/Data
M: Mixed

PROPRIETARY AND CONFIDENTIAL

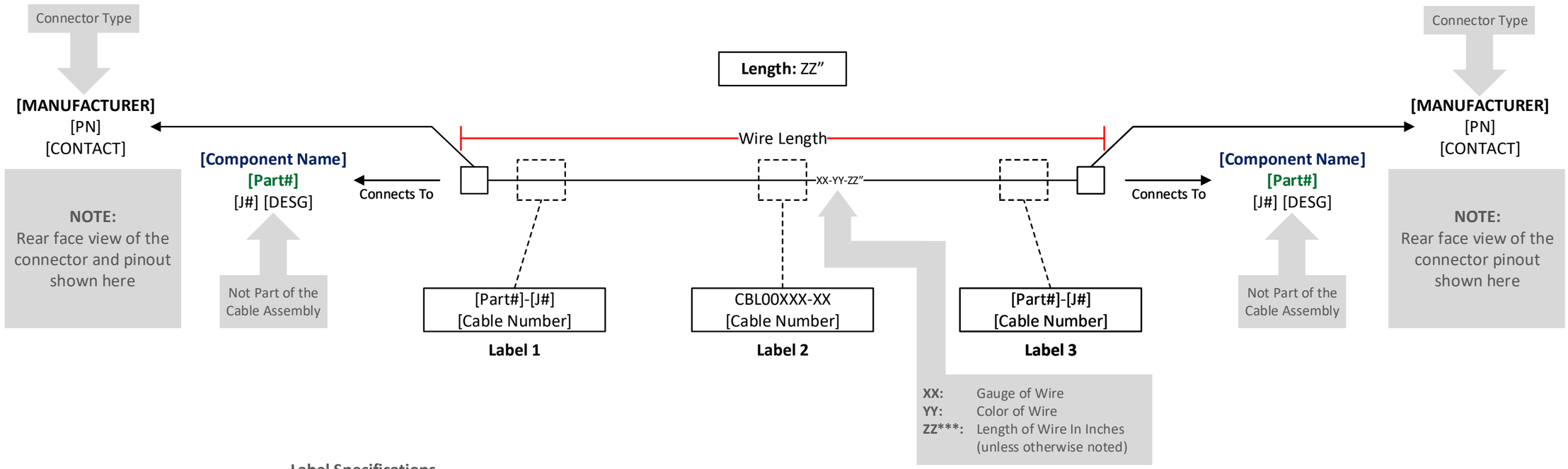
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UNCLASSIFIED

-	-	DRAWN BY	DATE	CACI EVER VIGILANT — 45200 Business Court, Dulles, VA 20166
-	-	S. MANNING	10/15/2019	
-	-	CHECKED BY	DATE	TITLE
-	-	C. BURGIN	10/16/2019	CABLE DRAWING, UNIVERSAL CHASSIS, YELLOWSTONE
-	-	ENGINEER	DATE	
ASY00478	YELLOWSTONE	C. BURGIN	10/16/2019	SIZE: B; CAGE CODE: 1KYV6; DRAWING NUMBER: CBL00040-01; REV: B
USED ON	PROJECT ASY	APPROVED BY	DATE	Sheet 1 of 12
APPLICATION		A. NICHOLS	10/16/2019	

UNCLASSIFIED

Cable Assembly Specifications



Label Specifications

Unless otherwise noted, following specifications must be followed.
**If cable length is 6" or less, only add one label (Label 2) in the center.

Type:	Heat Shrink (White)	**Label Location** (unless otherwise noted)**:	
Text Color:	Black	**Label 1:**	1" from the end
Background Color:	White	**Label 2:**	In the Center
Font Size:	As Needed	**Label 3:**	1” from the end
Side Margin:	.1"		

Length Specifications

- Length of the wire is from the tip of one connector to the tip of the other connector.
- Cable lengths under 6" can have a tolerance of +/- 0.25"
- Cable lengths above 6" can have a tolerance of +/- 0.5"

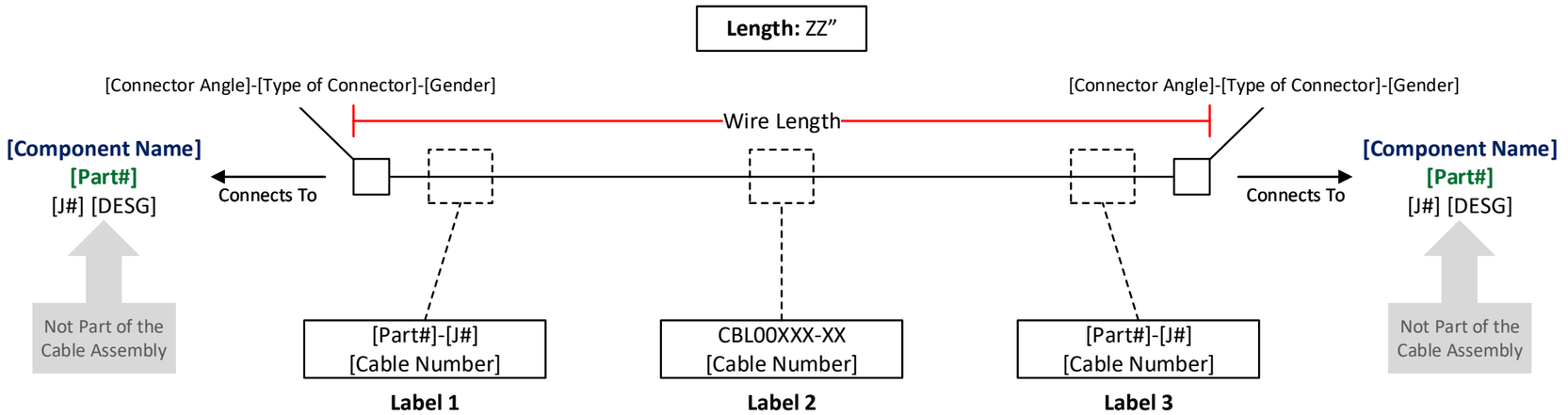
FOR PRODUCTION TEAM: Heat shrink **MUST** be used to cover the solder connections.

Example of Bill Of Materials

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1		(P/C/M)XXXX	CONNECTOR, Description	Manufacturer Part Number	Manufacturer Name		each
2			CONNECTOR, Description	Manufacturer Part Number	Manufacturer Name		each
3			CONTACT, Description	Manufacturer Part Number	Manufacturer Name		each
4			WIRE, Description	Generic	Generic		inch
5			WIRE, Description	Generic	Generic		inch

SIZE	CAGE CODE	DRAWING NUMBER	REV
B	1KYV6	CBL00040-01	B

RF Cable Assembly Specifications



Label Specifications

Unless otherwise noted, following specifications must be followed.
**If cable length is 6" or less, only add one label (Label 2) in the center.

Type:	Heat Shrink (White)	**Label Location** (unless otherwise noted)**:	
Text Color:	Black	**Label 1:**	1" from the end
Background Color:	White	**Label 2:**	In the Center
Font Size:	As Needed	**Label 3:**	1” from the end
Side Margin:	.1"		

Length Specifications

- Length of the wire is from the tip of one connector to the tip of the other connector.
- Cable lengths under 6" can have a tolerance of +/- 0.1"
- Cable lengths above 6" can have a tolerance of +/- 0.25"

FOR PRODUCTION TEAM: Heat shrink **MUST** be used to cover the solder connections.

Example of Bill Of Materials

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1		RXXXX	CONNECTOR, Description	Manufacturer Part Number	Manufacturer Name		each
2			CONNECTOR, Description	Manufacturer Part Number	Manufacturer Name		each
4			CABLE, Description	Generic	Generic		inch

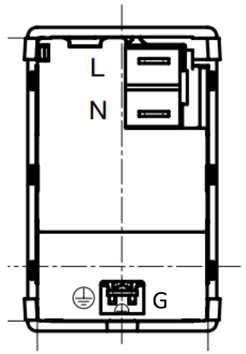
AC Power Entry

P0001

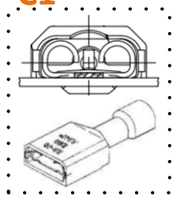
Length: Varies

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

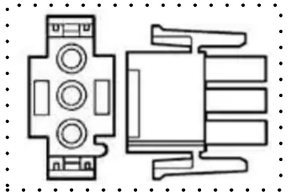
Module, AC Power Entry



C1

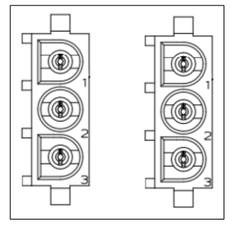


C2



FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

MOV Board



IN OUT

Module, AC Power Entry
Schurter
4301.1401, 0034.3126, KMF1.1191.11
J1

Connects To

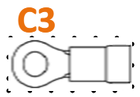
L	+120V AC	16-BK-6"
G	GND	
N	RTN	16-WT-6"

Connects To

+120V AC	1	
GND	2	
RTN	3	

MOV Board
CACI PCB00075-01
J1

C3



Chassis Ground Lug

16-GN/YL-7"

Chassis Ground
P0001

Label 3B

AC Power Entry
P0001

Label 1

CBL00040-01
P0001

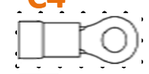
Label 2

MOV Board
P0001

Label 3A

16-GN/YL-7"

C4



Connects To

Chassis Ground Lug

Chassis Ground
P0001

Label 3C

Item	Ref/DES #	Cable Name	Description	Part Number	Manufacturer	Qty/Length	Unit
1	C1	P0001	CONNECTOR, Quick Connect, RCPT 16-14 AWG 0.250	3-350819-2	TE Connectivity	3	each
2	C2		CONNECTOR, Plug 3 Positions, 94V-0 UNI-MATE	350766-1	TE Connectivity	1	each
3	C2		CONTACT, Pin, 14-20 AWG, Crimp, Gold	350687-7	TE Connectivity	3	each
4	C3, C4		CONNETOR, Circular 16-14 AWG #10 Crimp	7113K24	Master-Carr	2	each
5			WIRE, Black, 16 AWG	See Specifications	-	6	inch
6			WIRE, White, 16 AWG	See Specifications	-	6	inch
7			WIRE, Green and Yellow, 16 AWG	See Specifications	-	14	inch

SIZE	CAGE CODE	DRAWING NUMBER	REV
B	1KYV6	CBL00040-01	B

Sheet 4 of 12

UNCLASSIFIED

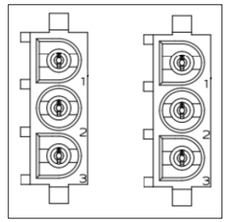
MOV Board to AC Input Module

P0002

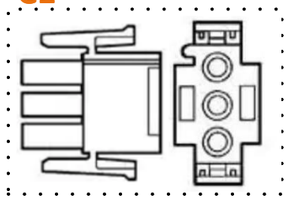
Length: Varies

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

MOV Board



C1



C2



x2

C3



MOV Board
CACI PCB00075-01
J2

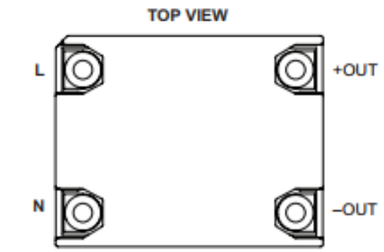
Connects To

1	+120V AC	16-BK-5"
2	GND	
3	RTN	16-WT-5"

+120V AC	L	Connects To
RTN	N	

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

Universal AC Input Module



Universal AC Input Module
Vicor AIM1714VB6MC7D5C00

Chassis Ground Lug ← 16-GN/YL-7"

Chassis Ground
P0002
Label 3B

MOV Board
P0002
Label 1

CBL00040-01
P0002
Label 2

Universal AC Input Module
P0002
Label 3A

Item	Ref/DES #	Cable Name	Description	Part Number	Manufacturer	Qty/Length	Unit
1	C1	P0002	CONNECTOR, Plug, 3 Positions, 94V-0, UNI-MATE	350766-1	TE Connectivity	1	each
2	C1		CONTACT, Pin, 14-20 AWG Crimp Gold	350687-7	TE Connectivity	3	each
3	C2		CONNECTOR, Circular, 16-14 AWG #6 Crimp	7113K23	McMaster-Carr	2	each
4	C3		CONNECTOR, Circular, 16-14 AWG #10 Crimp	7113K24	McMaster-Carr	1	each
5			WIRE, Black, 16 AWG	See Specifications	-	5	inch
6			WIRE, White, 16 AWG	See Specifications	-	5	inch
7			WIRE, Green and Yellow, 16 AWG	See Specifications	-	7	inch

SIZE	CAGE CODE	DRAWING NUMBER	REV
B	1KYV6	CBL00040-01	B

Sheet 5 of 12

UNCLASSIFIED

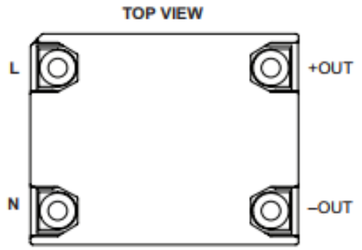
AC Input Module to AC-DC Converter

P0003

Length: 10"

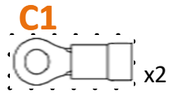
FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

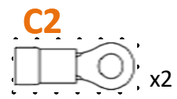
Universal AC Input Module

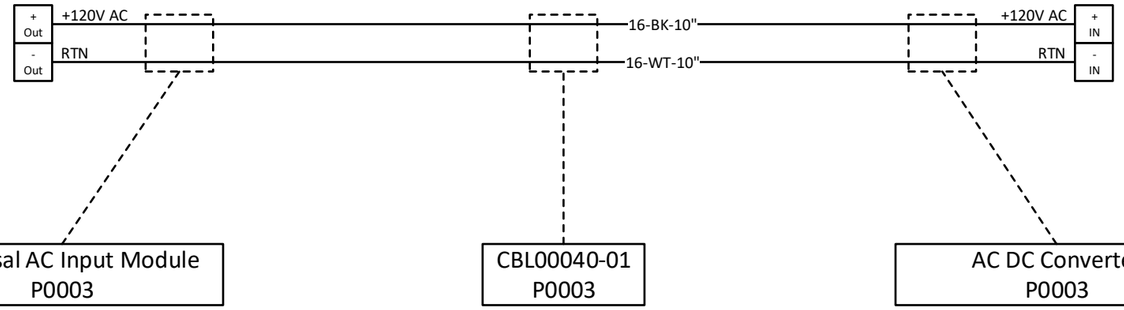


Universal AC Input Module
Vicor AIM1714VB6MC7D5C00

Connects To







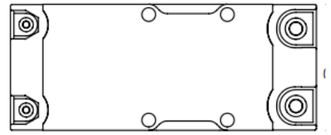
Connects To

AC DC Converter
Vicor PFM4414VB6M24D0CA0

IN

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

AC DC Converter



Item	Ref/DES #	Cable Name	Description	Part Number	Manufacturer	Qty/Length	Unit
1	C1, C2	P0003	CONNECTOR, Circular 16-14 AWG, #6 Crimp	7113K23	McMaster-Carr	4	each
2			WIRE, Black, 16 AWG	See Specifications	-	10	inch
3			WIRE, White, 16 AWG	See Specifications	-	10	inch

SIZE	CAGE CODE	DRAWING NUMBER	REV
B	1KYV6	CBL00040-01	B

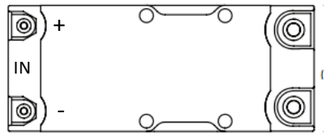
Sheet 6 of 12

AC-DC Converter to Capacitors and RF/Power Distribution Board

P0004

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

AC DC Converter



FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

Capacitor 1 & 2



Capacitor 1
KEMET ALS70A163DE063

Connects To

Length: 7"

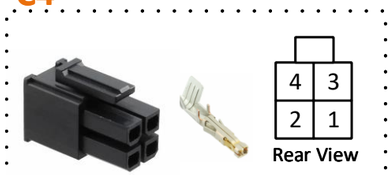
Capacitor 2
KEMET ALS70A243DF063

Connects To

Length: 7"

Length: 7"

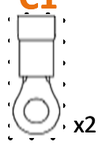
C4



AC DC Converter
Vicor
PFM4414VB6M24D0CA0

+24V DC
RTN
16-BL-7"
16-BK-7"

C1

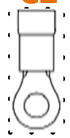


x2

CBL00040-01
P0004

Label 2A

C2



x2

CBL00040-01
P0004

Label 2B

C3



x2

CBL00040-01
P0004

Label 2C

1
2
3
4

Chassis RF/Power Dist. PCB
CACI PCB00064-01
VIN

Item	Ref/DES #	Cable Name	Description	Part Number	Manufacturer	Qty/Length	Unit
1	C1	P0004	CONNECTOR, Cicular 16-14 AWG #6 Crimp	7113K23	McMaster-Carr	2	each
2	C2, C3		CONNECTOR, Circular 16-14 AWG #10 Crimp	7113K24	McMaster-Carr	4	each
3	C4		CONTACT, Socket, 14-16 AWG, Crimp, Gold	1720630311	Molex	4	each
4	C4		CONNECTOR, Receptacle Housing, 4 Positions, 5.7mm	1716920104	Molex	1	each
5			WIRE, Blue, 16 AWG	See Specifications	-	21	inch
6			WIRE, Black, 16 AWG	See Specifications	-	21	inch

SIZE	CAGE CODE	DRAWING NUMBER	REV
B	1KYV6	CBL00040-01	B

Sheet 7 of 12

Fan and LED

P0005

Length: Varies

C1



2
1

Rear View

Chassis RF/Power Dist. PCB
CACI PCB00064-01
J5

Connects To

1 +24V DC
2 RTN

22-BL-9"
22-BK-9"

Chassis RF/Power Dist. PCB – J5
P0005
Label 1

CBL00040-01
P0005
Label 2

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY

*Fan wires are included with the fans

22-BL-15" +24V +
GND -

22-BK-15"

Fan 1
Sanyo Denki
109P0424J3023

22-BL-15" +24V +
GND -

22-BK-15"

Fan 2
Sanyo Denki
109P0424J3023



22-BL-15" +24V +
22-BK-15" GND -

Solders to

LED 24V Green
Dialight
6091222134F

LED 24V Green
P0005
Label 3

FOR REFERENCE ONLY;
NOT PART OF CABLE ASSEMBLY



Solder splice and heat shrink

Item	Ref/DES #	Cable Name	Description	Part Number	Manufacturer	Qty/Length	Unit
1	C1	P0005	CONNECTOR, Receptacle Housing, 2 Positions, 3mm	43025-0200	Molex	1	each
2	C1		CONTACT, Socket, 20-24 AWG, Crimp, Gold	43030-0008	Molex	2	each
3			WIRE, Red and Black, Twsited Pair, 22 AWG	See Specifications	-	54	inch

SIZE	CAGE CODE	DRAWING NUMBER	REV
B	1KYV6	CBL00040-01	B

Sheet 8 of 12

10 MHz IN

R0001

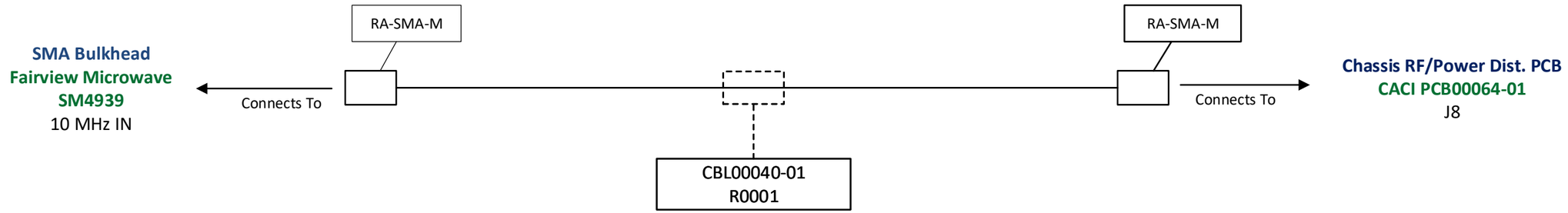
Length: 6"

C1



C2





Label 2

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1, C2	R0001	CONNECTOR, SMA Connector Plug, Male Pin, Right Anlge	AL20	KOAXIS	2	each
2			CABLE, Flexible, Low-Loss, Communications Coax	KF086	KOAXIS	6	inch

UNCLASSIFIED

10 MHz OUT

R0002

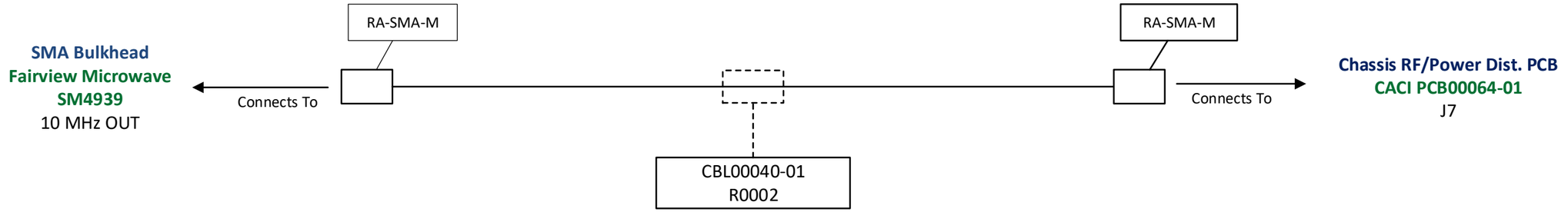
Length: 6"

C1



C2





Label 2

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1, C2	R0002	CONNECTOR, SMA Connector Plug, Male Pin, Right Anlge	AL20	KOAXIS	2	each
2			CABLE, Flexible, Low-Loss, Communications Coax	KF086	KOAXIS	6	inch

UNCLASSIFIED

SIZE	CAGE CODE	DRAWING NUMBER	REV
B	1KYV6	CBL00040-01	B

1PPS IN

R0003

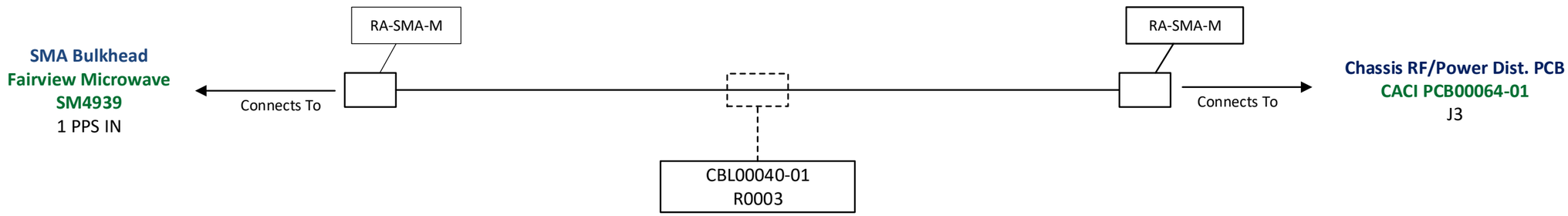
Length: 6"

C1



C2





Label 2

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1, C2	R0003	CONNECTOR, SMA Connector Plug, Male Pin, Right Anlge	AL20	KOAXIS	2	each
2			CABLE, Flexible, Low-Loss, Communications Coax	KF086	KOAXIS	6	inch

SIZE	CAGE CODE	DRAWING NUMBER	REV
B	1KYV6	CBL00040-01	B

1PPS OUT

R0004

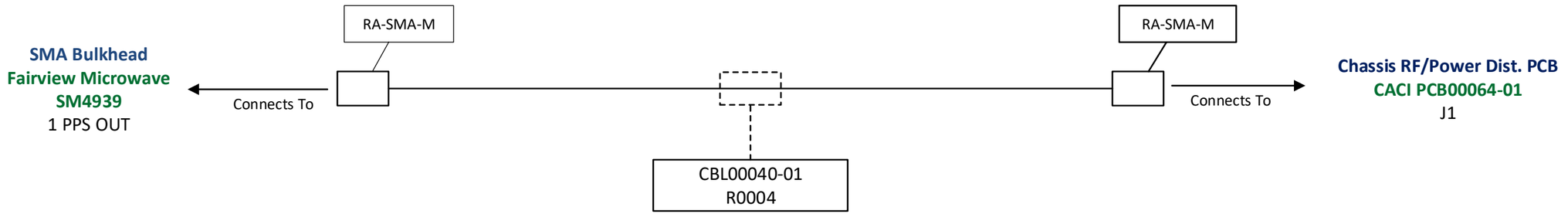
Length: 6"

C1



C2





Label 2

Item	Ref/DES#	Cable Name	Description	Part Number	Manufacturer	QTY/Length	Unit
1	C1, C2	R0004	CONNECTOR, SMA Connector Plug, Male Pin, Right Anlge	AL20	KOAXIS	2	each
2			CABLE, Flexible, Low-Loss, Communications Coax	KF086	KOAXIS	6	inch

SIZE	CAGE CODE	DRAWING NUMBER	REV
B	1KYV6	CBL00040-01	B

Sheet 12 of 12